

ActInflab 2021 Quarterly Roundtable #3

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Duration: 1:10:47 | **Uploaded:** Unknown **Transcript method:** auto_caption

hello everyone welcome back to the active inference lab welcome to our second semester today it's august 3rd 2021 so it's two cubed i guess nice date stephen please don't double or mute um welcome everyone we're really happy that you're joining us today in the active inference lab we're a participatory online lab that is communicating learning and practicing applied active inference you can find us at the links that are here on this page this is recorded in an archived live stream so please provide us with feedback so that we can be improving on our work all backgrounds and perspectives are welcome here and we'll be following video etiquette for live streams with muting if there's noise in the background and raising our hand when we're ready to speak today in our third quarterly round table we're going to go over the goals and provide sort of an overview on the lab itself then we're going to go into updates for each of the three organizational units of the lab which are the dot edu dot coms and dot tools and uh give some updates as well as talk about how people can get involved with each of those sections of the lab so we'll start with our goals maybe while i do some little tweak on the stream alex do you want to go ahead on this website slide uh no please go ahead oh sure well this is our website activeinference.org and if you go there now you'll see it looking much like this as well as providing a lot more information as far as links to follow us on and forms you can fill out but the goal of active inference lab is to produce cutting edge research and enable real-world applications of active inference so that's our inference side and our action side we're learning by doing and we're doing it about active inference so if you want to read more information about that check out the website to provide kind of a architectural overlay for the active inference lab this figure shows oh let me uh yeah this figure shows how the different organizational units that are internal to our lab are related to the interfaces and to external communities so i'll give an overview and then anyone can raise their hand and give a thought on it the three organizational units within the lab that we're going to be talking more about today are dot edu dot tools and dot coms and that's our internal states then at the interfaces that's where we have interactions with external entities so for example this live stream is a production of dot-coms it's right on the interface between the internal and the external states of

our lab and it interfaces out with broader communities so those are people who are learning active inference or curious about applying active inference or just what it is as well as all kinds of communities that may or may not realize that they're adjacent to active inference dot coms also reaches out to other groups and explores how active inference might play a role in their work and in edu the production on the interface is really the maintenance of the body of knowledge which again we'll be talking more about soon but we take a sort of systems engineering approach to diagramming our own activities and also to working internally and externally any thoughts on that yeah scott yeah i think the the value of structuring the uh research enterprise and the interaction enterprise in a way that reflects the content is absolutely superb that's something that um living the and breathing the active inference i think is really going to be instrumental as the i guess i'll call it the movement grows there's a lot of desire for deep risking out there and there's a lot of appeal in different areas banking food system security a lot of areas were starting to introduce the concept and having us organize the lab in that way um really walking the walk i think is is really terrific and speaks to the um flexibility and strength of the framing in a bunch of two different organizational constructs so i think that's something that's from the start was very wise for the org for you guys to do as an organization nice thanks it's almost like if you understand this org chart or this layout then you're you're half the way there you understand there's internal and external states and that they together so any other comments or we can continue on to the updates yes stephen well i suppose the only challenge you the the perennial challenge is what's internal what's external where's the boundary um and i think that that's proving to be an evolving conversation um i think that i think that's never answered in a way in a in a complete sense but um you know is it is it going to be more like there's multiple ways of thinking about that if you're doing statistics if you're doing some teleological morphogenesis type approach if you're doing some sort of top-down kind of using self-reported data and you're making a whole load of assumptions so i think that's quite quite quite useful and one thing that came up and i'm not sure which paper it was but i saw um carl first and i think he mentioned the idea of something called an approximation science which i hadn't heard before but there's almost a science of making approximations seem to come into play here um i haven't delved into that but that seems to also maybe tie into that question cool indeed another yeah yeah go ahead scott i just had another comment on stephen's comment you know that idea the boundary is really interesting because now because the lab uses the structure we'll be able to identify those parts of the lab function that are generative and the ways in which they're generative and so that

notion stephen about the boundary and where that resides gets really interesting then because the entire framing generates awareness and so the boundary in a way it almost is less important not that the question's not the question super interesting but the actual location of a boundary is almost less important than the processes that occur around a given arbitrary boundary within because system is going to have multiple boundaries and so it's a fascinating question of the where the round a boundary resides kind of within a complex of boundaries all of which are engaged in active inference simultaneously in the system because it's one of the things i'm understanding is there's no mono dimensional system that's beyond a theoretical system your their systems are running all sorts of active inference calculations simultaneously of different weightings etc and so the boundary starts to become like drawing a circle with tangents in a way drawing a boundary with tangential active inference uh dynamics in a way and and you have multiple active interfaces going to it so that was a fascinating the question you just asked got me thinking about that um that could boundary construction in a different way and very dynamic boundaries that would occur thanks nice and uh yep many ways of going about active inference as we'll be discovering and we arbitrage that against the clarity of what active inference lab is itself like each of these units are their own informational blanket there's a drive folder there's a weekly meeting so we're sure about that those are not the ambiguous boundaries but that's actually how we go about investigating the real world and the systems that do have uncertainty that's fundamental let's go on to the updates so the first update which happened right at the end of last semester was that actin flab has a new steady state as a non-profit organization registered in california united states of america and this was part of our labs strategy and medium-term plan right from the beginning any thoughts on this alex or we'll have more to say on this in just a few slides um about this action and these new states important thing important thing to think about is the new affordances and again we can consider it in a notion of active inference what possible opportunities we can get and you use for strategy realization some new contacts and i think we have some uh information in slides about on some next slide about what possible opportunities are yes one of the cornerstones and motivators of our project and lab as well as the registration as a non-profit especially is this concept of research debt and alex how about you maybe summarize what research debt means in this context [Music] in general the notion of research depth uh is about some needed uh and special actions what need to be done for some new state-of-the-art research uh because people who are doing uh really great research in their fields spending very time to get to wet frontiers but after that to force such fresh knowledge to be accessible to provide the audiences in

different domains some additional works need to be arranged and execute and there are some practices what we understand uh what need to play their role and make methodological work methodical work and our work with some of our disciplines to make this knowledge accessible for anyone with any level of expertise in the field yes it's a technical topic that we're studying and so there's a lot of work that goes into getting a working understanding and so how are we going to scale that and make that accessible stephen yeah i think one other aspect is trust you know there's um they say in community development you you can well this one approach i really like is that you move at the speed of trust i think that's quite a nice thing and a non-profit shows a sort of commitment to values which you know implicitly is kind of there in an academic organization but the thing with academic organizations there's certain limitations in that scope where a non-profit space does offer a sort of a more pluralistic flexible space as well but again making sure that you can establish that trust in the values which i think the commercial sector um even with people like google who said a lot about the values early on then it starts to you know so i think that is an important feature that you know is part of this process cool thank you how does the non-profit registration update the phenotype of our lab well again using active inference as a framework for understanding this we can break it down in terms of the affordances that were provided as well as the new niches which can be physical economic informational the new niches that we're able to access so we're better able to scaffold our long-term partnerships we're able to engage in new kinds of interactions like collaboration with other legally recognized entities or support of various kinds will be able to apply to new grant programs as lead or as support will also be able potentially support lab employees in a sort of annualized way or in a project-based way and it will overall just help us serve through deep time as we provide improved onboarding and accessibility efforts so anyone who's interested in this set of fields or any other field that they think is going to be relevant anything related to non-profit management or lab management research program management making grant applications some of the more classical areas of interest for non-profits that are research and technological as well as some of the more novel or online characteristics that are important like crypto economics nfts uh crowdfunding token engineering graphic design website development tool development we'll talk more about some of these things later and a special note for this second semester and then anyone can provide a thought is uh we're planning a lab advisory board we're planning to arrange this lab advisory board to start operations during this second semester so the rest of 2021 and we welcome the suggestion or invitation to individuals with high expertise in their domain people who would

like to be part of our and some of the following areas of the lab activities will be supervised and catalyzed by the board especially legal education governance methodology research strategy application slash innovation and really anything else so this again helps us anchor out by connecting with external stakeholders and providing people who maybe they're not going to come to the dot edu meeting every week but maybe once a quarter the uh the planets align and we're able to interface with this person and so that can be hopefully a win-win interaction so if you're listening and you think this is you or you know somebody who would be cool to invite onto the board just let us know about that any thoughts on this or we'll continue on to the organizational units yep scott so two thoughts just on the exempt uh status it's great congratulations on that one is that the advisory board can also be a source of funding and other resources and network obviously and so constructing an advisory board and looking for those folks i think that my impression is again from the conversation i'm having with finance folks etc is that there's going to be a lot of interest uh people are looking at the framing and i think getting quite excited about it so there may be a lot of action there and you can have different advisory boards multiple boards as just as a structural matter just with different functions etc and you can also one of the things to think about in the beginning is sometimes having the advisory board be a two-year stint or a three-year stint or something so you can naturally roll people off if people kind of linger too long or if you want to freshen it up so kind of establishing that at the beginning is something to think about keep it fresh um one other thing if the um if your exempt organization purpose is includes educational then there i recall that there's authority out there in private letter rulings where educational institutions were permitted to engage in standard setting which gets really interesting one of the things i've told people involved in education i just spent time with some k-12 folks last week is that when there are standards out there in the world education is about teaching the standards so if i go to learn french or chemistry i learn the standards of french and chemistry in school where there are no standards then the standard setting is done by curriculum and so one of the things this teaching that you're doing here is also standard setting because you're teaching framings where in spaces where there may be none and so it's something to think about that the function of this lab may be both educational and standard setting and just to be aware that a lot of the outputs may feed into standards efforts as these there could be standards of business operations law technology and society that can come out of this work and so that's something to be aware of and to be intentional about if we start to go in that route and making sure that the deliverables that are intended to be standards have the maturity and the um exposure so that they can have

a chance as standards it's something there's a real opportunity in these new spaces these new framings to pursue that as well and under your auspices of your 501 exemption thanks scott very nice stephen yeah just further into the the the finding standards and also looking for processes where you've got developmental aspects and that's so this might be where there's a kind of a couple of streams going on because um in a similar way to a lot of non-profits that work in highly innovative developmental spaces you've got elements of that here unusually for um because it's not just the material within a very standard framing the vehicles being developed um at the same time so developmental evaluative approaches principles focused evaluative approaches could be um which are actually very new in the non-profit world which have taken that have grown massively might really be good for certain aspects but there may be other aspects which start to get locked down in more ready already established processes and so that could be really really useful and we'll come back to this idea of standards in the next section which is the dot edu so it's our first organizational unit because learning is really at the heart of the actin flab in edu the goal is to create a participatory and dynamic active inference body of knowledge and this is really drawing on alex and yvonne's experience with the systems engineering body of knowledge which isn't just a reference book it's not just a wiki it's not just the certifications not just the standardization it's actually this way of thinking about how these pieces interact and support everything from career development to people who are just curious to conferences all these different moving parts that play a role in that living body of knowledge so we'll provide an update here on the main efforts of edu which are the ontology project with a working group as well as a uh an ontology in progress document and then we'll talk a little bit about the next steps but any overall thoughts alex on what body of knowledge means or how edu helps us get there i also want to highlight that a lot educational activities in priority and we understand that only doing some projects with some new people in the field could lead us uh to have broad community people who can work in a project with active inference that's why we are trying to approach this very fundamentally and go in through his hard way through methodological ontological works and always wet hard stuff yep totally agreed so here's uh yeah steven go ahead yeah one part that i think is really valuable with the the body of knowledge is this idea of the system of interest which i know alex is you know they've got a really well-defined and evolving approach of using systems of interest and how that interfaces or with this active inference approach to a bounded system or object and i think that has proved to be sort of a consistency which can bring things into the real world in these participatory ways beyond say purely modeling in a more academic context

so i think that's been consistent and is is a certain rigor that opens a lot of good questions but i i'm really excited by that here's a figure that we've seen several times it's a spectrum that is describing a uh sort of from the least formalized forms of ontology on the left side to the most formalized forms the most expressive informal forms of ontology on the right side and knowing that ontology which is how the world is studying the world and how we frame it for active inference we knew that was gonna be a challenge but really the hill that we had to climb and we started in the first quarter of right when our lab begun with compiling just a terms list just what terms were relevant what were we seeing come up what were terms that people were thinking were relevant for understanding papers and we're going to be working our way towards the right side of this continuum by introducing increasing levels of annotation and formalization right now what we're working on is updating our terms lists which we'll show in the next slide as well as introducing references and definitions so first just where have other people defined a term in text and then eventually we'll move towards creating our own synthetic definitions and then also introducing translations and translations into natural human languages as well as translations eventually into sulo sumo suo kif which is if people want to check that out that's really what the ontology working group is doing they're this train scooting to the right day by day uh trying to understand this best-in-class way of working with ontologies that adam peace and others have developed and so we're working through the textbook the ontology textbook which everybody's welcome to learn by doing as we work through this textbook and figure out what sumo is and we'll use that to start to develop computational ways of thinking about the terms that are relevant and along the way we're going to be structuring and providing all kinds of interesting ways of augmenting those lists and just show what it looks like and then anyone can provide a comment and especially dave who's done a ton of awesome work on these uh tabs in our working ontology document which we'll share with anyone who wants to see it there's a few tabs there's the core terms which are split into a definitions and references tab where we get up to five references so published papers that say you know term x is this and just quote them what was it described as in the text which isn't always the clearest or isn't always the most consistent across definitions but that's the point we need to know that and then the second tab we have translations and at this point we have russian portuguese spanish french and italian are mostly or completely done but if you speak any other language it'd be awesome to have your translation effort then we have a terms supplement where we have kind of a bunch of other terms that we think might be relevant or might get upgraded to a core term later or just are good to keep on hand in this

broad dictionary that we're building out as well as in equations tab which dave did some nice copying over of equations because even the equations the formalisms have have somewhat evolved or a letter will mean something in one field but then it means something different in another field or in a later paper so we need these kinds of formal ontologies because otherwise somebody will say oh but g meant this in this other paper and it may have meant that so how do we go from an internally coherent like within the paper coherent formalism to usages and equations that are consistent across so that's what this work of connecting the dots reading between the citations that's what the ontology working group is really working on okay and the so again if you're interested in this at all come contact us and join our ontology working group the other dimension or project within the dot edu is this instructional and interaction design and the big question here is in the medium term how are we going to provide the affordance for people to learn active inference to update their generative model through action and inference about what active inference is and how to apply it and we're especially excited about this dialectic between instructionism and interactionism and how that might relate to course design and we talked about this in active livestream 23 and dean and others had a lot of really interesting things to say about this but suffice to say that instructionism is the idea that you're going to just pass the telegram and it's going to be received as an instruction carried out like a cookbook interactionism is sort of where active inference takes us which is yes there is information passing between the entities however it's not simply that the telegram message is being received and the recipe is being carried out there's something ongoing that's dynamic that's being re-normalized at every instance to for example the surprise that the sensory unit is experiencing or the signals that the proprioceptive aspects of the body are sending back so what does that look like instead of just making a sort of list of terms or things you're going to walk through and then after you've gotten all these instructions you'll know active inference how could it be something that's co-active with multi-agent sense making and people working together and interacting to actually improve their model and then just the last question and then stephen is how do we provide applied active inference education for individuals coming from just so many different backgrounds that's going to be a question we're always focused on stephen it might be as well when you look at that interactionism it might well be that when people use or learn about how to interact with how much of the active inference or what kind of model um would they maybe know about so that that sort of raises that question of the different people's need to know so to speak of what active influence represents or is modelling could vary as we enter into some fields of systems engineering some fields of psychology

some fields of uh so that could be quite interesting is what kind of complexity or scope of the model is needed to onboard for a particular sort of use case or discipline thanks scott you know it got me thinking um the last two you're coming and stephen's comment you know um the active inference the framing it allows a new matured perspective on how information works and one of the things we were on i was going to call earlier today with a group that just put together a group field guide to misinformation narrative campaign field guide it's called and one of the things we were talking about was that there's a sophistication we're kind of wiring up the collective human mind now and active inference is kind of a necessary prerequisite to being a good neuron or a good participant neurons the wrong word certainly um a good participant in that system and there's kind of a civic duty to kind of understanding states and understanding your state and other states and i won't use the word empathy or compassion but there's definitely an intentionality and an awareness that active inference lends to systems understanding of how communication works and because folks who i i s asserting these days that the mind doesn't exist in the brain it exists in language and in culture and the brain is just tuned into the local mind so we tune you know when you're babies you that's why colonial powers want to have indigenous children in their schools they want to tune their minds to imperialism so one of the things that's kind of interesting is we're constantly mind tuning each other and the awareness of other people's internal states and the fact of these lends a new power and also new responsibility to shared communications and as the species is getting tighter packed and becoming in the world is becoming smaller informationally there's almost a responsibility and duties associated with an active inference framing and it's kind of a respectful element to it or an awareness an interdependence element to it recognizing um this space and what is going on in space and and just to that point when i first came into the space the first thing i noticed was i said oh my god my whole life all i've been doing is contracts which are synthetic markov blankets right and it just there's awareness that i had of what i was doing in the world and how i could frame it in the context of active inference that really does um lead to affordances social affordances where there's a co-awareness of the processes taking place that can absolutely offer superior pathways to governance because information governance can be risk governance if properly approached and so this kind of framing can lead and carry the burdens of that kind of thing because it's accessible enough once the basics are understood and if it's a shared framing it becomes an interoperable way of being and that's part of what's missing in information now is interoperability um on cross-cultural basis thanks thanks scott so kind of to close the section and then stephen any last comments on edu

everyone is welcome to get involved with this organizational unit through the project-based learning and the big umbrella that everything is under is knowledge engineering so whether you're translating languages human languages computer languages whether you're in the archive in stacks looking for definitions and references if you're working on curriculum development and what that looks like taking into account that whole interesting dialectic of instructionism and interactionism or whether you're in the ontology working group learning ontology which is just such a vast area but you're going to be learning with other people who are also learning whichever of these affordances you take hopefully it'll be something that you can uh get a lot out of by working with a group stephen yeah building just one last point and tied to what scott said is states the idea of states um and i think that's actually proved to be where some of the last of the uh terms list was being sort of worked on as things that related to states is a very much a jump from what is being done in most fields so you have even in systems it's often what's been moved around the dynamics the the type of objects but the state of the object is not normally exactly featured in the same way and it's the same even in psychology someone's state might change but often it says well they've been triggered it's almost easier to say there's a trigger and there's a thing that's triggered someone and talk about it in terms of objects and stuff then actually the state has changed so um i think this is this is quite a hard concept to get a head round but comes up time and time again and of course inactive we partition states in a model dependent way into internal and external and then blanket states or boundary states which first and then subdivided into action and sense but there we go we'll talk more about that in the coming weeks the next organizational unit is dot coms which is communication and dot coms has an internal and an external goal and function the internal goal is to organize all of the labs internal projects and activities that's kind of an organizer role and then the external goal is to carry out all forms of communication with external entities so if you email activeinference@gmail.com or if you contact us on twitter whatever it happens to be that interfacing externally is a dot-coms topic we're going to talk in these next few slides about some of the follow-ups from last semester and then we're going to look forward at a few of the papers we've selected to read in the coming weeks as well as give opportunities for places where people can make suggestions for who'd be an awesome guest or what papers would be great to read so the first follow-up and uh dave if you want to add any comments on this you'd be more than welcome on june 21st near the end of last semester we had our first applied active inference symposium with carl fristen and we had three different sessions each about an hour long carl was just such an incredible guest and really generous

with his time and his answers super educational and you can check out the bitly link or you can go to our youtube page where you'll find these videos and in terms of a follow-up something that dave suggested and took the initiative on is to generate a transcript of the proceedings of this symposium and then we'll curate that add in the references clean it up and then hopefully pending everyone's approval have that transcript as a citable preprint so instead of oh there was this one video that one time where i think this person said this but maybe i'm not remembering it or how many times does somebody say that and it's like a tenth of them will go to the timestamp and then a tenth will really listen to what it says well we're going to have a text representation that we can then translate we can connect back to the formal ontology and we can just contribute to this knowledge accumulation and especially some of his answers which are not written out anywhere else it'd be really unique and important to have the information in this way so that's the follow-up from the fristen symposium this short link here which i'll also post into the live chat is the short link where you can always go to understand what is going to be happening in the upcoming weeks with the streams in terms of the papers that we're reading so here we are um in the beginning of august and in the first two weeks of august we're going to be reading this recent paper bayesian mechanics for stationary that's a pretty technical mathematical paper and then in the following weeks we're gonna um have authors on discussing everything from industrial engineering to non-equilibrium thermodynamics in late october and we also haven't yet selected the papers for 28 29 and 30. so three papers that we're going to be able to choose so if anybody has a paper that they would like to read we have a list of unplaced readings in this document so you can scan through that list but if you have another paper you'd like to suggest we're also always open to and especially if it's your paper or if you can contact the author and get them to join us for those weeks then that's bonus points and this first tab of the spreadsheet is tuesday at 7 pacific or 1400 gmt it's a participatory group discussion so it's best if you've at least tried to read the paper in terms of scanning through it but no need to fully understand it no need to be an expert we're all just going there and helping each other do sense making on these really interesting papers so please feel free to get involved if you're curious about live streaming or learning science out in the open however it presents itself to you hopefully this is an affordance for you to get involved with live streams and with actinflab so that's our usual tuesday live stream series so at this same exact time that we're doing this live stream except other than the four round table meetings in the year every other week we do a journal discussion dave yeah have we ever looked at a paper by martin butz uh in germany uh toward a sub-symbolic theory of

computation i'm really enjoying that i don't think so let's send you the link yeah great cool so as far as what are the affordances when we want to discuss a cool paper so some papers we bring on to this tuesday live stream we're going to unpack it for two weeks in a group hopefully with the authors and we're also going to do a zero video as a lab where we unpack the paper and kind of give this uh summary and also background and context on the paper but we also have the affordances of these other series of streams and just to kind of give a flavor and also get people excited for what is upcoming let's talk about a few guest streams and model streams that we have coming up so this is on august 9th just next monday we're going to have mark miller is the primary presenter along with julian kiverstein ines hippolyto and lars sanfed smith the guest stream is going to be called exploring the predictive dynamics of happiness and well-being and just to read the first sentence of their abstract in this talk i will present some of our recent work on mental health and well-being using the active inference framework i i want to read more but you're going to have to go listen to this guest stream because i know it's going to be a great one so that's an example of a guest stream is some authors want to present on ongoing or recent work or they want to just raise up some perspective and we can just schedule any time that works for them you know 48 hours later or three months later the guest stream is kind of our wild card where any thing that somebody wants to present that's related to active inference we're always happy to give the venue for and another upcoming event this is going to be on august 19th so just in two weeks is going to be model stream number 4.1 with bert devrie this von der lar and dimitri bagaiv and it's going to be implementing active inference by message passing in a factor graph and this is related to the work of the bias lab which is producing this forney lab julia toolbox for factor graft based probabilistic programming so this diagram on the upper right is related to what they're going to be talking about so if you see that you think that looks so cool or how is that related to active inference this is a really powerful tool that these fellows are working on so we're all excited to learn more about that and maybe implement it in some cases so that'll be a great model stream that's just some of the kinds of papers as well as guest appearances that we get to feature if you want to give a guest stream on your research or if you have a suggestion for somebody who would be great for a guest stream or math stream or model stream we're always open to that and bonus points if you can reach out to the author or if you are the author so that we can know that it's going to actually happen and right now we're planning any kind of event and presentation for the rest of 2021 you know if it is a date in 2022 that you need to set aside let's do that too but we want to make the rest of 2021 powerful and really accessible

to a lot of different backgrounds so if you're interested in doing a guest stream or suggesting just get in contact with us anytime we'll also in this dot coms section give a shout out to cognitio 2021 the website is here this is going to be an online conference in the middle of october so only about a couple of weeks away and there's great co-organizers they just updated the website to reflect some of the speakers and also several of our lab participants are going to be presenting research at this conference so if you're curious about active inference and collective intelligence and also a lot of social topics this is going to be an exciting conference so just to close out in dot coms and then anyone can provide any thoughts there are many affordances for participating in dot coms we welcome people who have been involved in the lab already or who want to get involved with leading and growing the actin flap accounts on social media people who want to contribute or improve the state of our current series which are the podcast which is primarily blue's work as well as the live streams anyone who's interested in introducing new series and formats in compiling and curating the research products of the active lab participants we've already had multiple conference presentations papers where actinlab was written as an affiliation for an author people who want to help onboard new participants in live streams so they understand how to set up everything from their audio and visual to conceptually what should one do to prepare designing events blog short writing tutorials anything else that strikes your fancy so hopefully if you're excited about communication and or active inference one of these opportunities is right for you any last comments on comms before we go to dot tools all right our third organizational unit of three is dot tools the goals of dot tools are to uh again kind of like how dot coms had an internal and external facing side dot tools has sort of a a current and a future facing side our current facing side is to enable effective tool use and instrument use for all active processes so whether it's the video chat platform we're using or live streaming those are tools that we're using so we want to use them skillfully but also a little bit more broadly dot tools is interested in exploring and designing affordances for our niche resulting in effective action as well as innovations in tool development and dot tools last semester had many weekly meetings where resources were shared and a lot of needs and ideas related to the systems engineering of tools were discussed to kind of give the overview this is a figure from our 2020 paper active inference and behavior engineering for teams where we framed the operations of a remote team in terms of the structure of active inference models which is these multi-scale nested markov blankets with sense coming into the agent and action going out to an agent and communication is kind of this special case whether it's human communication or two apis communicating or a human in

a server because one's action is the other one's sensory state and so that kind of focus allows us to take a function driven approach for how we think about communication and tool development using active inference so i know that in tools brainstorming group people were coming up with just many different avenues and getting a lay of the land in terms of like what does an active inference tool look like does it look like a news feed but the ranking algorithm is different or does it look like a physical object does it look like a new way of interacting with an object we already have a lot of pieces there any other thoughts on this before um alex why don't you then give a overview of the net hack challenge uh scott you had a comment oh hey i just wanted to mention briefly on the tools one of the things that's really interesting to me and new to the active inference world here is that um the notion of how the tools reveal differentials and my work i was an attorney for three decades and worked in risk basically that's what attorneys do we try to de-risk things and it usually has to do with differentials information differentials understanding differentials and i think one of the things is going to be really interesting is as the tools develop how might they be used in risk analyses in different areas i think that there's the the ability to navigate and close differentials that's implicit in the active inference framing i think is going to prove to be exceptionally valuable in industries that deal with risk finance and other institutional construction so it's just something i wanted to raise about the tools that right now they're used it seems like their analysis of active inference itself but i think they're incidental benefits of these tools they'll find application in myriad domains outside of academic interest in academia where the academic interest in active inference where there's interest in de-risking and the tools will be used directly to facilitate the closing of gaps of information that are associated with risk thanks yep very cool one of the applied projects oh yeah stephen go ahead well i was just going to say in in terms of the developmental nature as well i think you know the um in instructionalism and um interactionalism which has been the developing theme i think you could say in the lab over the last few months starts to now maybe revisit some of those earlier papers you might say okay so we've got action so we've got the idea of information being passed through or imprinted into a blanket as kind of blanket states which goes into become internal states and you've got the blanket somehow you know impression in itself on the environment so you've got this this question about um okay so you traditionally we may have just been looking more instructionally as in information comes in information goes out well now we're looking at this more interactionist dynamic with the system boundary you've both got an information flow and some sort of action the actors doing it and that that then also becomes interesting maybe

as we look at tools this sort of question comes up um and even how to diagram that is up for grabs a little bit at the moment but uh i just thought that really made me think of that cool scott yeah just one other point and i think there's a lot of movement now where from a graph theory perspective the difference of nodes and edges we spent a lot of human history concentrating on nodes and that's really associated with physical space it feels like the edges the interactions the relationships and the ability for us to describe attributes of edges and relationships i think is increasing and it's increasingly part of how we do identity and other things that we rely upon in moving through space and de-risking and leveraging our behaviors so that stephen's coming and that notion of tools really um highlighting the space between nodes and really allowing us to start to parse relationships and parse the nature of interactions and the nature of state changes and things like that what's really going on dynamically and the bottom line is i think it gives us a lot more solution space with within which to think together um and so that parsing of interactions allows us to alight in that solution in middle space between nodes and to explore together with other people especially when there's awareness of and so it's very similar to what's happening now i think in the word misinformation is beginning to attenuate and people are starting to realize that we just have a lot of information out there and there's all sorts of context and content and intention and all sorts of things so we can't just say misinformation or and similarly the relationships need parsing the subtlety and nature of the interactions of different nodes in systems i think is going to be revealed increasingly by awareness of the attributes of the relationships that are going to be re helped to be revealed through active inference analysis thanks cool all right um alex on to the net hack challenge so all well and good but then how do we really apply these techniques and how will we give a example that amazes people what do you think about this alex and what's the vision here yeah thanks this challenge is looks really interesting just because of its nature about that legendary game and uh a lot of people with different ages backgrounds know something or maybe even play from one side from another side it's really one of the difficulties most difficult games in terms of solved by computers by machine learning techniques so and also for now it's interesting that challenge is hosted on the on research conference and so there are some connections between possibly smart agents in this game and connections with neuroscience and in terms of complexity and the visibility and we just have to try to show up some real life examples of applying active inference agents to some tasks because even in our streams we saw some examples of real life agents which operates in complex environment so we know we have proofs that it could be possible on that technical examples so we are trying to invite

everyone from the community who are interested in machine learning techniques and active inference approach trying to solve this task i don't know how really hard it will be but at least it should be fun and could be first real life applications for with programming code of active inference models scott i have a question um when the when active inference was first kind of being explored and proposed my understanding was and tell me if this is accurate that it was being abs proposed as a model for modeling biological and cognitive systems things that were extent in the environment and see if that's the case one of the things i'm wondering about is what are the differences when we intentionally invoke and apply the model in a system versus when we observe it in an existing system are there differences can we mimic is it biomimicry for us to pick up this framing and use it and what are the challenges we and alex question to you in your experience so far in kind of designing this where we're desiring to create and and this goes back to my comment about contracts being synthetic markov blankets we are creating i never thought of them as that but it was if i were thinking of him is that it might have made me do better contracts so what are some of the differences if there are any between observing an active inference framing and creating an active inference framing in a system you want to give a thought alex um yeah i have a lot of questions around this too but uh watching latest model streams on our channel uh brings more perspectives and understanding what need to be touched uh to do it it's it's almost like when we're talking about a system that we're making measurements from we're doing inference on the dynamics of a system in the world and then when we it does rely on eventually measurements and inference to make sure everything is going down the way that we want it to but actually it's a creative action that we're undertaking to truly design something and so it's a interesting question especially in the purely software realm do we just port over the same exact um sort of spatial biological or sensory affordances or are there new ways that a biological system wouldn't have access to like long-range communication or some type of internal model tweak that just doesn't really exist in biological so great question and uh we'll have the nethack challenge team come back to give you some thoughts later dave oh you're muted dave on the other side okay i still can't hear you but when you figure it out we'll come back to you but yeah um that's the net hack so it's happening from uh right now until october 15th so if you want to get involved in nethack challenge this would be the right time to do it and the skills as laid out in our discord and also on that previous slide especially if you're interested in learning and applying machine learning hacking gaming all these skills are important this is not like tetris or pac-man the game itself is actually uh something that has to be played through because there's it's a challenging game and

um it should tell people something that go chess starcraft these games have been played to extremely high level by just off-the-shelf neural network algorithms and nethack hasn't and so that's sort of the barking dog you know the dog that didn't bark it should let people know that it's the type of problem maybe that integrates multi-level planning or symbolic or discrete elements those are the types of problems that yes we can recognize the dog in the image but can we play nethack and the answer is no also there's a special prize category for people who are using a non-neural network so all things to think about um also other affordances in dot tools uh people who are interested in applying active inference through product research and engineering and development that's an availability alex will also be uh helping us through a systems engineering project based learning process and of course anyone who wants to help us develop and improve our tool usage any thoughts on the systems uh side alex okay um um okay now dave now dave yeah can you hear me now yep yep oh good okay uh yeah my actually my my major question is whether anything is going on with rosetta hub have you heard from tim at all it seemed like he vanished he's in the game he's in the game you got your commute because you're on a serious feedback on a serious feedback loop say that again i had to mute dave just because he was on a serious feedback loop um with rosetta and whatever else we use for cloud architecture for the ontology working group will keep people informed on those channels scott one of the things i think is kind of interesting here is the if and to the extent that um active inference is something that's naturally occurs and we can observe in systems whether they know they're doing it or not which i i assert to people i say that you're doing active inference whether you know it or not and there's articles on how humans use a beijing knowledge construction so those beijing notions are kind of embedded in a lot of human and other activities to that extent it's kind of interesting because we i always thought about this as bringing the active inference framing to the stakeholders if and to the extent they're already doing active inference it's kind of becomes a coaching element also of revealing the skills that are already there the things that are already being done by the system and then um tweaking them to maximize the throughput of whatever is desired by that organization so it's a i hadn't thought of before is active inference as a naturally occurring phenomenon as an end as a synthetic phenomenon and we use the word artificial but it's something we synthesize and how what we're doing may be discovering an improvement of an existing human affordance and and not just human but living and that puts a really different tweak on it it's it's like um you know you are here in a map you know telling someone they're already doing and let us show you how to improve it it's like a capacity

mining in a way and and something we might think about in the tools because i hadn't thought about it that way before but again the people doing it and observing it where it occurs among insects and cells which don't vote and you know don't send out memos and advocating i'm advocating at least for among humans who do vote and trip over their own feet emotionally and politically all the time and so it's kind of interesting if active inference is a um can be demonstrated to be an efficient and or resource useful or some other positive qualities because nature engaging in it and that's a demonstration of it it offers an interesting rhetoric to its instantiation in synthetic context as well that hey you're already half you're already doing it why not do it well kind of idea something to think about cool scott thanks steven just to let you know i've i have heard from tim in the sense that he is going to be sharing something on a session uh soon he's just had a few uh there's a few delays but um we've got the support from uh karen and latifah and uh so i think progress will be shared shortly cool great um all right let's look towards the future in our last little section the next steps for the lab which are participatory so it's not going to happen without people who want to step up and step in and make it happen people who want to act the next steps are described within each unit so for edu comms and tools we talked what kinds of projects we're going to be doing this semester and hopefully some of the affordances if you want to help contribute or get involved there we're always looking to onboard new participants so if you're just hearing about this and it's always the right time to get so contact us and then as we head into our second semester of activity we really want to focus more on defining our governance structures and on exploring these new affordances that are available to the non-profit when we were working with the terms list and then there were these interesting discussions like how will we decide what is a core term and yes there's the conceptual question of what constitutes a core term but then there's the social question about not everyone's going to have the same exact perspective on which terms should or shouldn't be in the core terms and maybe nine out of 10 people just don't care and one person decides to say yes or no on it but that's not a governance structure so how are we going to be able to work truly as a collective in a participatory in terms of the products that we make and the process how we get there along the way so those will be exciting steps for our second semester and we'll kind of just close out with some of the questions that we always ask about papers but our system of interest is the lab as organizers and as participants so we want to know the same exact things that we would ask about a paper like what do we want to learn how do we want to apply it what would we gain or what would be different if we had a good understanding what are the unique predictions and implications how are we going to check up

along the way what are the next steps what is our policy selection how are we going to reduce uncertainty about the ways that we achieve our preferences what are the goals of the research and then that remainder the modulo which is our curiosity and how are we going to be driven by that differential to keep on minimizing our free energy so that we can continue developing the lab scott one of the things for us to be aware of is we're the lab is a machine to build machines in a way right we're teaching the the lab is by doing the thing you're teaching or learning or exploring one thing to be really careful of is when people come in that they don't get confused by that layering this is it happens a lot i'm building a lot right now of governance systems to build governance systems and so it's just something to be aware of that when we're describing the functions of the lab that we're really explicit about the it being an instance of the thing it's developing as well because that can get confusing even though it's kind of there's a naturalness to it you're living it's like gandhi you know be the future you want to see kind of thing but or whatever the quote was but it's something to be aware of in terms of the description of the undertaking that we have this yes we're doing active inference and yes we're living active inference so when sometimes when we're talking about active inference stuff we're talking about the organization itself and sometimes we're talking about active infant stuff we're talking about the generalized concept of active inference and for us to they're not they don't need to be separated heavily but they do need to be juggled because there is a layering that's a that's natural to us because we're working in it now but a new entrant will get because they'll be confused on what lev at what level they're engaging at certain times so we need to be very explicit about the which level of recursivity we're existing in when we're dealing with something in those multiple levels cool very true so people can give any last thoughts just raise their hand but otherwise we'll just say thanks so much for participating this was a great start to our second semester of activities in the lab any feedback or suggestions or questions that people have we're always open and the links here contain how people can stay informationally updated stephen and then anyone else that might be a nice linking to the next uh live stream because i think that will talk about um what it means to segregate something as an object and when it means to maybe do more you know predictive processing or and then when you segregate things into active and sensory states so i think i think that is an interesting question that as we get more immersed will be quite interesting is at what point does someone mostly need to know that there's something happening at a very embodied action sensory process or where is it like the starting point is just that there's predictive processing going on of some sort and that it's not all purely information coming in and then we

have the whole question about system of interest again which is always fun so thanks oh thank you any other last comments uh yeah scott i just wanted to thank all of you for being so generous with your time and your knowledge and your ability and willingness to engage to spread this this to me is a very valuable and um type of set of insights and it allows for a great deal of new innovation in a lot of different areas it goes to fundamentally to how we think and so i want to thank you for being beautiful people and beautiful in terms of your generosity and your wisdom so it really is a pleasure to be able to associate with you and learn from all of you so that's something i look forward to in the next semester thank you thank you scott been great times so onto the paper let's get on that bayesian mechanics paper and start reading it and highlighting every word we don't understand yet aka all the words all the equations but um thanks everybody for participating live and for watching we'll see you in the coming weeks with the guest stream with mark miller at all on monday and then following we'll just be back in the rhythm of the papers alright see you later